

Some basics of integration

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 - Properties and useful techniques.

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 - Informal review of the basics. Functions of one variable.
 - The fundamental theorem of calculus and some implications.
 - Properties and useful techniques.
 - Integration and differential equations; an example.

Anti-differentiation and indefinite integrals

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- $F(x) + c$ is an anti-derivative (or indefinite integral) and c is the constant of integration.
- In terms of notation, we write

$$\int f(x)dx = F(x) + c.$$

to say that we are looking for the anti-derivative.

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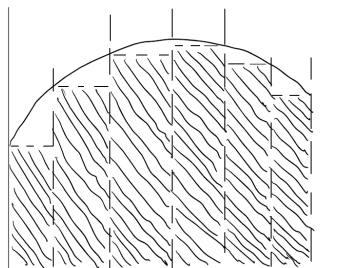
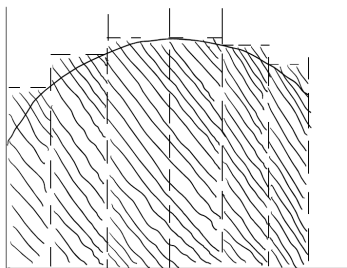
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- But, there are many ways to approximate $f(x)$ on each sub-interval.

Definite integrals

- Two approximations (over- or under-estimates):



Definite integrals

- We are calculating the area of a number of rectangles and summing them up. With n sub-intervals, we can write our estimate as

$$\sum_{i=1}^n f(x_i) \Delta_i, \quad (1)$$

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- Comparing (1) and (2): You can think of $\int$ as meaning “summation” and of dx as an “infinitesimal small” change in x .

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- That is, c does not matter when we calculate the area under $f(x)$.
- The fundamental theorem of calculus is useful for many reasons and in many applications, such as welfare evaluations (equivalent and compensating variations).

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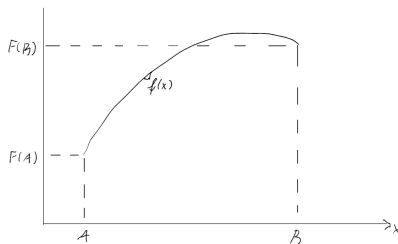
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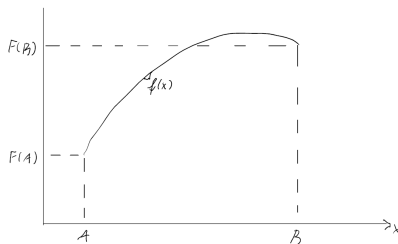


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- Example: A firm's cost function can be inferred from knowing its fixed costs and its marginal cost function.

An aside on strictly increasing functions

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- Example: $F(x) = x^3$ is clearly a strictly increasing function. Indeed, $f(x) = 3x^2 > 0$ when $x \neq 0$, yet it holds that $f(0) = 0$.
- Implication: “ $F'(x) = f(x) > 0$ for all x ” is a stronger statement than “ $F(x)$ is strictly increasing”. The latter does not imply the former.

Properties/techniques I: Addition

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- Intuition: The area over $[a, c]$ can be calculated as the sum of the area over $[a, b]$ and $[b, c]$.
- Notation is often abused: When we are integrating over the entire domain, it is common to just write $\int f(x) dx$. If we are not integrating over the entire domain, then we need to clarify what the interval is.

Properties/techniques II: Reversing the direction of integration

Reversing the direction of integration changes the sign of the integral,

$$\int_a^b f(x) dx = - \int_b^a f(x) dx.$$

Properties/techniques III: Integration by parts

Consider the product of two functions $G(x)f(x)$ with $g(x) = G'(x)$ and $f(x) = F'(x)$. An extremely useful rule is then that

$$\begin{aligned}\int_a^b G(x)f(x)dx &= [G(x)F(x)]_a^b - \int_a^b g(x)F(x)dx \\ &= G(b)F(b) - G(a)F(a) - \int_a^b g(x)F(x)dx.\end{aligned}$$

Properties/techniques IV: Integration by substitution or change of variables

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- Instead of thinking of x as the variable, we can think of $u = g(x)$ as the variable. Note that the term $g'(x)dx$ equals du . Hence, we obtain

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du,$$

where we adjusted the limits of integration to reflect that u takes different values than x .

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- *Leibniz's rule* says that

$$\begin{aligned} \frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx &= f(b(t), t)b'(t) - f(a(t), t)a'(t) \\ &\quad + \int_{a(t)}^{b(t)} \frac{\partial f(x, t)}{\partial t} dx. \end{aligned}$$

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 - Step 1: Derive the expenditure function from observed demand.
 - Step 2: Use the expenditure function to infer or recover preferences.

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 - The solutions to the two problems coincide: $h_i(p, u) = x_i(p, e(p, u))$.
 - Shephard's lemma tells us that $\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u)$.

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- Thought-experiment: Income, w^0 , and the price of good 2, p_2^0 , stay fixed. We observe changes in the price of p_1 , and observe how demand change. Thus, we think of $e(p, u)$ as a function only of p_1 .

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- Since only p_1 changes, this tells us that

$$\frac{de}{dp_1} = \frac{e}{2p_1} \iff \frac{de}{e} = \frac{1}{2} \frac{dp_1}{p_1}.$$

Differential equations

- Example: *Unbeknownst* to the outside observer, the true utility function is $u(x_1, x_2) = x_1 x_2$.
- The Marshallian demand function for good 1 is

$$x_1(p, w) = \frac{w}{2p_1}.$$

- As p_1 changes, we know from (3) that

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- Since only p_1 changes, this tells us that

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- We now integrate – or find the antiderivative – on both sides

$$\ln e + c = \frac{1}{2} \ln p_1 + c'.$$

Differential equations

- Starting from

$$\ln e + c = \frac{1}{2} \ln p_1 + c',$$

rearrange to get

$$\begin{aligned}\ln e &= \ln (p_1)^{\frac{1}{2}} + c' - c \\ &= \ln \sqrt{p_1} + \ln \exp(c' - c) \text{ where } \exp \text{ refers to Euler's number} \\ &= \ln \sqrt{p_1} + \ln k, \text{ where } k = \exp(c' - c) > 0 \\ &= \ln k \sqrt{p_1}.\end{aligned}$$

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- Thus, we conclude that the expenditure function takes the form

$$e(p_1, p_2^0, u^0) = k \sqrt{p_1},$$

as a function of p_1 . However, k is still unknown.

Differential equations

- To find k , we use

$$e(p_1, p_2^0, u^0) = k\sqrt{p_1},$$

and the initial condition, (4), that

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- In conclusion,

$$e(p_1, p_2^0, u^0) = \frac{w^0}{\sqrt{p_1^0}}\sqrt{p_1} = w^0\sqrt{\frac{p_1}{p_1^0}}.$$

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- Combining,

$$e(p_1, p_2^0, u^0) = 2\sqrt{\left(\frac{w^0}{2p_1^0} \frac{w^0}{2p_2^0}\right) p_1 p_2^0} = w^0 \sqrt{\frac{p_1}{p_1^0}},$$

which is exactly what we found by integrating observed demand!

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- Step 2 uses the expenditure function to infer or recover preferences.

Recovering preferences from the expenditure function

- We will consider the boundary of the set

$$\begin{aligned} V &= \{x \in \mathbb{R}_+^2 \mid p_1 x_1 + p_2^0 x_2 \geq e(p_1, p_2^0, u^0) \text{ for all } p_1 > 0\} \\ &= \{x \in \mathbb{R}_+^2 \mid p_1 x_1 + p_2^0 x_2 \geq w^0 \sqrt{\frac{p_1}{p_1^0}} \text{ for all } p_1 > 0\}. \end{aligned}$$

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- If $x \in V$ then the bundle always costs at least as much as the cheapest bundle that achieves u^0 , *regardless* of what p_1 is.

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- If $x \in V$ then the bundle always costs at least as much as the cheapest bundle that achieves u^0 , *regardless* of what p_1 is.
- If $x \notin V$, then there is some p_1 for which x is strictly cheaper than the cheapest bundle that gives utility u^0 . Thus, x is an *inferior* bundle, because otherwise the expenditure minimization problem was not solved correctly in the first place.

Recovering preferences from the expenditure function

- For any given p_1 , the constraint in the set is satisfied if

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- Substituting into the constraint and simplifying yields

$$x_1 x_2 \geq \frac{1}{4} (w^0)^2 \frac{1}{p_1^0 p_2^0}.$$

Recovering preferences from the expenditure function

- The boundary of the set is given by the (x_1, x_2) bundles that satisfy

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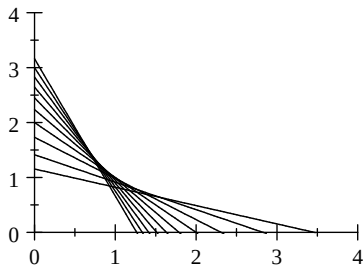
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- A graphical illustration on the next slide.

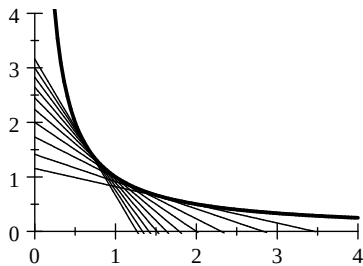
Recovering preferences from the expenditure function

We looked at the boundary of the set

$$V = \{x \in \mathbb{R}_+^2 \mid p_1 x_1 + p_2^0 x_2 \geq e(p_1, p_2^0, u^0) \text{ for all } p_1 > 0\}.$$



(a) $p \cdot x = e(p, u^0)$ for many p



(b) Superimposing the indiff. curve